

VASANTHA VEMULA

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SUMMARY

Full Stack .NET Developer with 5+ years of experience building enterprise applications using C#, ASP.NET Core, Angular, SQL Server, Microservices, Docker, and Kubernetes. Experienced in designing scalable REST APIs, optimizing application performance, modernizing legacy systems, and delivering production-ready solutions across finance, utilities, and industrial automation domains.

SKILLS

- **Languages & Frameworks:** C#, .NET Framework, .NET Core, ASP.NET MVC, ASP.NET Core MVC, ADO.NET, Entity Framework Core, Dapper
- **Backend & Architecture:** MVC Architecture, REST APIs, Web APIs, SOLID, DRY, YAGNI, Dependency Injection (DI), LINQ, Object-Oriented Programming (OOP), SOLID Principles, Design Patterns, Performance Tuning.
- **Frontend Technologies:** Angular, AngularJS, React, JavaScript, TypeScript, jQuery, AJAX, Razor Views, HTML5, CSS3, Bootstrap
- **Cloud & DevOps:** Microservices, Azure DevOps, Azure App Service, Git, Docker, Kubernetes, IIS, Jenkins, CI/CD Pipelines,
- **Databases:** SQL Server, Oracle, MongoDB, T-SQL
- **Testing & Tools:** Selenium WebDriver, NUnit, MSTest, xUnit, Postman, Splunk, Swagger.
- **Reporting & Data:** SSRS, SSIS, SSMS, PDF Generation, Large Dataset Processing.
- **Core Areas:** Enterprise Application Development, Financial Systems, Production Support, SDLC, Agile/Scrum, Requirement Analysis
- **AI Tools:** GitHub Copilot, Claude, Gemini, ChatGPT

Professional Experience

Koryak Consulting Inc

01/06/2025 - Present

Client: Allegheny County Retirement Board (Sr. Software Developer)

- Designed and developed enterprise pension administration applications following the ASP.NET Core MVC architecture using C#, .NET Core, WinForms, Angular, JavaScript, TypeScript, Entity Framework Core, and SQL Server, supporting retirement processing, payroll, refunds, and benefit administration.
- Built Excel exports using stored procedures and Entity Framework for faster, optimized reporting.
- Created PDF documents from Word templates using Memory Stream for efficient in-memory file handling.
- Refactored service methods to improve modularity, clean architecture, and file storage processes.
- Fixed threading issues with Entity Framework DbContext, improving application reliability.
- Wrote clean, well-structured jQuery scripts to enhance front-end functionality and maintainability.
- Worked with complex View Models to streamline front-end and back-end integration in MVC applications.
- Implemented secure upload and download processes for file handling within the application.
- Designed modular application architecture following SOLID principles and enterprise design patterns.
- Engineered complex financial calculation modules including retirement estimates, payroll proration, tax calculations, payment allocations, and eligibility validations.
- Developed, modified, and optimized SSRS reports, datasets, shared data sources, stored procedures, views, and database objects to support operational reporting, business analytics, and enterprise reporting requirements across large datasets.
- Built and maintained automated CI/CD pipelines using Azure DevOps and Git.
- Optimized Entity Framework Core, SQL Server queries, and application performance supporting high-volume enterprise workloads.
- Implemented data persistence using Entity Framework Core with LINQ, repository patterns, and optimized database transactions.
- Integrated enterprise applications using HTTP, JSON, Swagger, and Postman to enable seamless communication between internal systems.
- Investigated and resolved complex production issues involving business logic, authentication, IIS, SQL Server, Entity

Framework Core, SSRS reports, deployments, and application performance through detailed root cause analysis.

- Created technical documentation, deployment documentation, and implementation guides throughout the SDLC.
- Leveraged GitHub Copilot and AI-assisted development tools to accelerate development and improve productivity.
- Collaborated with Business Analysts, QA teams, and stakeholders throughout Agile/Scrum development.

Client: Pittsburgh Water & Sewer Authority (PWSA)

- Developed enterprise ASP.NET Core MVC applications supporting GIS administration, work order management, inspections, and appointment scheduling.
- Developed interactive dashboards and data visualizations using D3.js, creating dynamic charts, graphs, and custom visual components for business reporting and analytics.
- Designed configurable exception processing, notification workflows, and administrative modules.
- Built REST APIs supporting enterprise integrations and customer service operations.
- Implemented server-side pagination, SQL optimization, and processing of large operational datasets.
- Supported production deployments, monitored application health, resolved production issues, and coordinated release management activities.
- Worked closely with business users during requirement gathering, technical solution design, and production support.

Client: Laborers Combined Funds

- Gathered project requirements for a new Laborers Combined Fund system, documenting use cases and workflows.

Fluid Management, Wheeling, Illinois

.Net Full Stack Developer

Oct 2023– Dec 2024

- Migrated legacy Java application modules to C#/.NET 7, modernizing the application architecture while preserving existing business functionality.
- Applied Dependency Injection with repository and service patterns to decouple business logic from data access and improve application flexibility.
- Designed scalable RESTful APIs and microservices supporting enterprise business workflows and distributed applications.
- Engineered a System on Module (SOM) solution to automate partition reset operations, eliminating storage-related update failures on embedded devices.
- Integrated IoT-enabled paint dispensing devices with backend services and REST APIs, enabling reliable device communication and real-time data synchronization.
- Split application features into independent microservices, making it easier to develop, test, and deploy changes without impacting other modules.
- Deployed and managed containerized applications on Kubernetes, including updating deployments, monitoring pod health, and troubleshooting failed pods
- Used Docker to containerize .NET applications, allowing developers to run the same application consistently across development, QA, and production environments.
- Deployed, scaled, and monitored Dockerized .NET microservices on Kubernetes using Deployments, Services, ConfigMaps, and Secrets.
- Collaborated with DevOps teams to deploy containerized applications through automated CI/CD pipelines using Azure DevOps.
- Investigated and resolved issues related to microservice communication, container failures, and deployment errors by analyzing application and Kubernetes logs.
- Optimized SQL Server database performance by developing stored procedures, views, and query tuning techniques while collaborating in Agile development teams using Git.

Logic Loops LLC, Alpharetta, Georgia

.Net Developer

Mar 2023 – Sep 2023

- Developed and maintained .NET applications, ensuring robust performance and scalability.
- Built and maintained web applications using ASP.NET Core and C#.
- Developed and debugged REST APIs supporting backend services and application integrations.

- Helped set up CI/CD pipelines using Azure DevOps to automate build and deployment.
- Used Git for source control and collaborated through pull requests.
- Containerized .NET applications using Docker to provide consistent development and testing environments.
- Assisted in monitoring applications using Azure tools like Application Insights.
- Wrote basic PowerShell scripts for automating repetitive tasks.
- Participated in Agile team meetings and sprint planning.
- Helped manage secrets and config settings using Azure Key Vault.
- Documented development processes and deployment steps for team use.

Tata Consultancy Services, /NAPA Auto Parts, Chennai

.Net Full Stack Developer

Nov 2020– Jan 2022

- Developed and enhanced enterprise web applications for NAPA Auto Parts using ASP.NET Core MVC, C#, Angular, Entity Framework Core, SQL Server, JavaScript, HTML, CSS, and Bootstrap, delivering responsive and user-friendly interfaces.
- Developed reusable Angular components, implemented client-side validations, and integrated REST APIs to deliver responsive enterprise web applications.
- Modernized legacy applications by migrating the Data Access Layer from ADO.NET to Entity Framework Core and optimizing SQL Server queries, improving application performance and maintainability.
- Developed RESTful APIs and integrated front-end components with .NET backend services while following MVC architecture, Agile methodologies, and SDLC best practices.
- Automated application deployments using Azure DevOps CI/CD pipelines, managed source code with Git, and collaborated with cross-functional teams to deliver production-ready releases and application enhancements.

THESPAN, India

.Net Full Stack Developer

Jan 2020 – Oct 2020

- Developed and maintained full-stack web applications using ASP.NET MVC, C#, React.js, Vue.js, JavaScript, Bootstrap, HTML, CSS, Dapper, and SQL Server, delivering responsive and user-friendly interfaces.
- Designed reusable UI components, integrated RESTful APIs with .NET backend services, and optimized SQL Server stored procedures, views, and complex queries to improve application performance and scalability.
- Developed automated test frameworks using Selenium WebDriver, C#, and Postman, integrating API and regression testing into Jenkins CI/CD pipelines to improve software quality and release efficiency.
- Supported cloud deployments using AWS services (EC2, S3) and Jenkins to streamline application delivery and continuous integration.
- Collaborated with cross-functional development and QA teams throughout the SDLC, contributing to application development, testing, defect resolution, and production-ready software delivery.

Education

Master's in computer science

New England College, New Hampshire Jan 2022 – July 2023

Bachelor's in information technology

B V Raju Institute of Technology, Narsapur, Hyderabad affiliated to JNTUH 2016 - 2020